Thomas Ford

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PROFESSIONAL SUMMARY

Senior AI Engineer with deep experience building and deploying Generative AI, LLM, and RAG solutions across healthcare operations, patient engagement, and clinical workflows. Skilled in Python, prompt engineering, agentic frameworks, GCP, and MLOps practices including MLFlow, Docker, and CI/CD pipelines. Introduced 6 LLM prototypes at CVS Health that cut investigation cycles by roughly 30%, and reduced patient-service processing time by 24% through AI-augmented API redesign. Brings a responsible-AI mindset and cross-functional track record translating complex business requirements into scalable, auditable GenAI systems.

TECHNICAL SKILLS

Backend: Python, Node.js, REST APIs, T-SQL, PySpark, Express.js, Event-Driven Architecture, LLM Orchestration

Frontend: React, TypeScript, Redux, Material UI, HTML5, CSS3, Accessible Forms, Responsive Web Applications

Cloud: GCP, Vertex AI, AWS, Docker, Kubernetes, GitHub Actions, CI/CD Pipelines, CloudWatch

Data: Snowflake, PostgreSQL, SQL Server, Redis, MLFlow, Data Modeling, SQL Query Optimization, Audit Logging

Tools: MLOps, Prompt Engineering, RAG, Agentic Frameworks, GitLab, Postman, Swagger/OpenAPI, Jest

Industry: Generative AI, LLM Security, Responsible AI, Value-Based Care, Revenue Cycle Management, HIPAA Compliance, EHR Integration, Clinical Workflow Automation

PROFESSIONAL EXPERIENCE

Senior AI Engineer | CVS Health
Remote | 05/2023 - Present

• Architected and deployed 6 production-ready Generative AI solutions using Python, Vertex AI, and GPT-based LLMs across 12 patient-facing workflows spanning scheduling, virtual care, prescription summarization, and benefits eligibility, collaborating with product, clinical, and compliance stakeholders.

• Increased appointment-flow completion by 31% by engineering LLM-powered registration guidance with prompt engineering and retrieval-augmented context injection across CVS Virtual Care patient journeys.

• Reduced patient-service processing time by 24% by deploying Python-based agentic workflows that unified scheduling, prescription, and benefits data retrieval behind a single orchestrated GenAI API layer with Snowflake-backed caching.

• Introduced 6 LLM prototypes for support summarization, automated Jest test generation, and workflow-gap analysis, cutting selected investigation cycles by roughly 30% for clinical operations and engineering teams.

• Stabilized GenAI-augmented care journeys at 99.94% availability by redesigning LLM output moderation, access controls, and CloudWatch alerting across containerized Vertex AI services.

• Accelerated model iteration cycles by 41% by standardizing prompt versioning, quantization workflows, and responsible-AI guardrails including output moderation and auditability controls aligned with HIPAA and PHI handling requirements.

Senior Software Engineer | BRIDGE Hospice

On-site | 02/2021 - 05/2023

• Led architecture for 9 patient-intake and care-coordination workflows using Python, React, C#/.NET, and SQL Server, delivering AI-augmented document summarization and clinical task-routing features in close coordination with nurses, physicians, intake coordinators, and billing stakeholders.

• Streamlined referral validation and intelligent task routing by deploying a Python-based classification model, cutting intake handoff time in half while preserving nurse review checkpoints and PHI-aligned audit controls.

• Modernized 14 React screens and backend endpoints with AI-assisted form pre-population, improving first-pass intake completion by 27% across clinical and administrative user cohorts.

• Reduced peak SQL Server response time by 38% through query restructuring, index tuning, and background-job scheduling optimized for care-coordination dashboard data retrieval.

• Eliminated 42% of repeat production defects by expanding xUnit, Postman integration, and Cypress test coverage across admission, document-processing, and provider-communication paths.

• Resolved one-third of admission-status escalations by exposing LLM-generated workflow-state summaries and operational alerts to coordinators, clinicians, and product owners through a unified REST API layer.

Software Engineer | UMC Physicians

On-site | 07/2018 - 12/2020

• Designed and delivered 8 MyTeamCare patient-portal capabilities across online scheduling, paperless registration, provider messaging, and medical-record access using JavaScript, React, REST APIs, and SQL, engaging physicians, front-desk staff, analysts, and infrastructure teams across 16 EHR data-mapping integrations.

• Transformed appointment booking into a single-session guided flow backed by eligibility-aware business logic, raising successful self-service scheduling by 33% without increasing call-center staffing.

• Simplified paperless registration forms through data-driven field validation and conditional rendering, reducing incomplete submissions by 22% while improving mobile accessibility scores.

• Implemented secure session management, authorization controls, and input validation that lowered account-access incidents by approximately 20% across patient and staff user populations.

• Revitalized 11 legacy portal modules using reusable React components and integration-test coverage, decreasing release-related regressions by 29% over two consecutive release cycles.

• Generated 13 operational and utilization reports backed by optimized SQL queries and targeted caching, shortening weekly data-reconciliation cycles by roughly 35% for analytics and operations stakeholders.

Junior Software Engineer | GermBlast

On-site | 07/2016 - 06/2018

• Developed 10 scheduling and field-operations modules using JavaScript, C#/.NET, MySQL, and responsive web forms for dispatchers, technicians, and customer-service staff, enforcing 15 validation and permission rules across customer, appointment, and service-record workflows.

• Improved dashboard load time by 41% by rewriting MySQL queries and eliminating unnecessary data retrieval across service-location views, directly reducing dispatcher response latency during peak scheduling windows.

• Solved recurring scheduling conflicts by implementing database constraints and deterministic assignment rules, lowering duplicate bookings by 26% over a 6-month measurement period.

• Expanded role-based permissions across 19 user profiles, cutting unauthorized record-change incidents to zero during the final year of the engagement.

• Generated 13 operational reports for invoicing, service history, and technician utilization that shortened weekly reconciliation cycles by roughly 35% for operations and finance stakeholders.

• Restructured data validation logic across customer and appointment workflows, reducing form-submission error rates by 19% and cutting operations team correction requests by more than one-third.

EDUCATION

Bachelor of Science in Computer Science | University of Kansas | Graduated May 2016